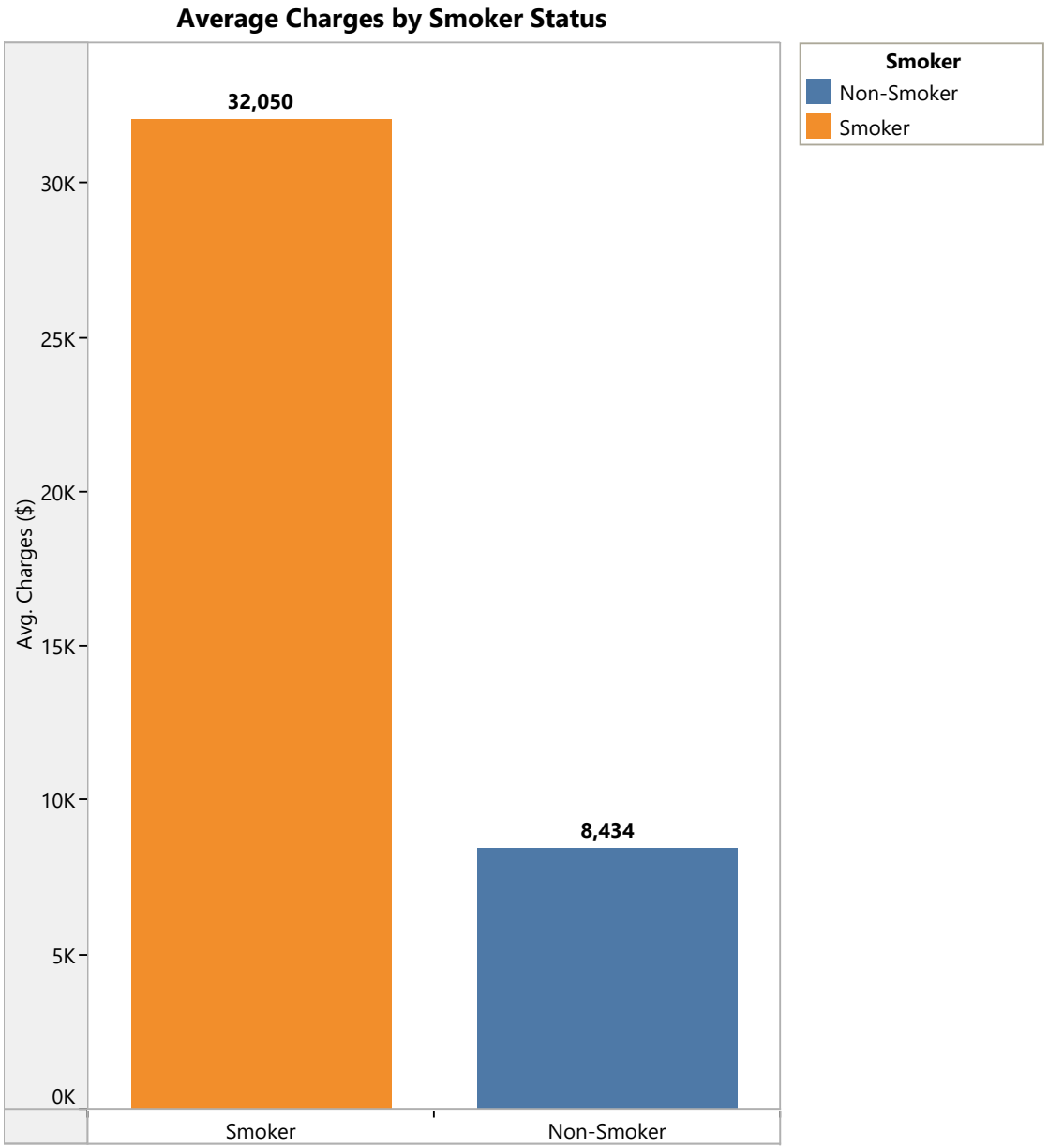
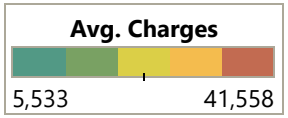


The Top 10 highest medical claims all come from smokers, with charges ranging from $47k to $63k. These extreme outliers are concentrated among obese smokers—especially in the Southeast and Southwest—confirming that smoking is the strongest cost driver in the dataset.

Smoking is the strongest cost driver in the dataset. Smokers consistently show more than double the average medical charges of non-smokers.

Smoker × BMI Category Heatmap

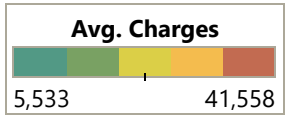
BMI Category	Non-Smoker
Underweight	5,533
Normal	7,686
Overweight	8,258
Obese	8,843



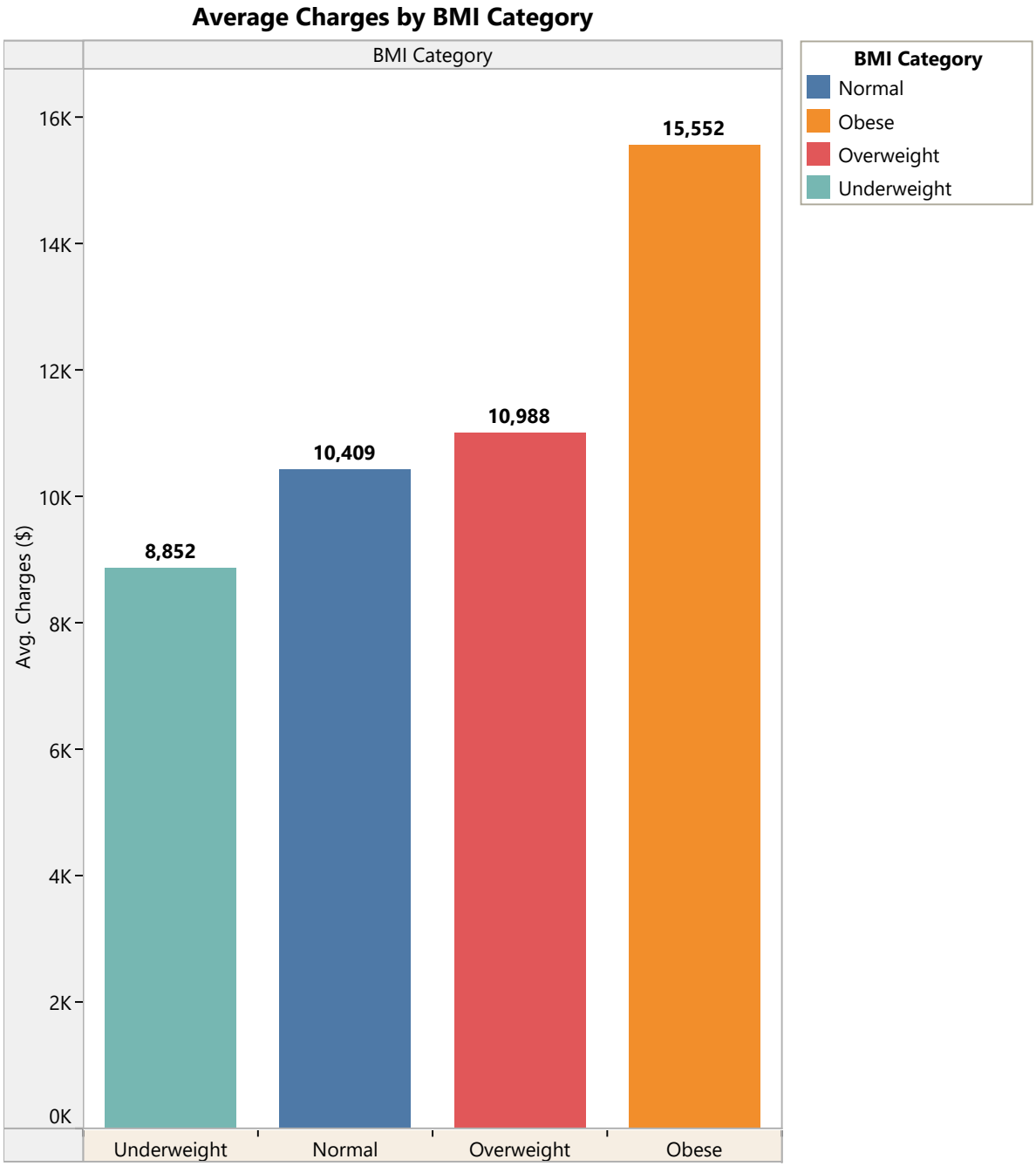
Smoking outweighs BMI as a cost driver. Underweight smokers have higher average charges than obese non-smokers, including a $32k claim. Obese smokers remain the highest-cost group overall.

Smoker × BMI Category Heatmap

BMI Category	Smoker
Underweight	18,810
Normal	19,942
Overweight	22,496
Obese	41,558



Smoking outweighs BMI as a cost driver. Underweight smokers have higher average charges than obese non-smokers, including a $32k claim. Obese smokers remain the highest-cost group overall.



Medical charges increase steadily with BMI. Obese individuals have the highest average costs due to elevated chronic disease risk and greater healthcare utilization.

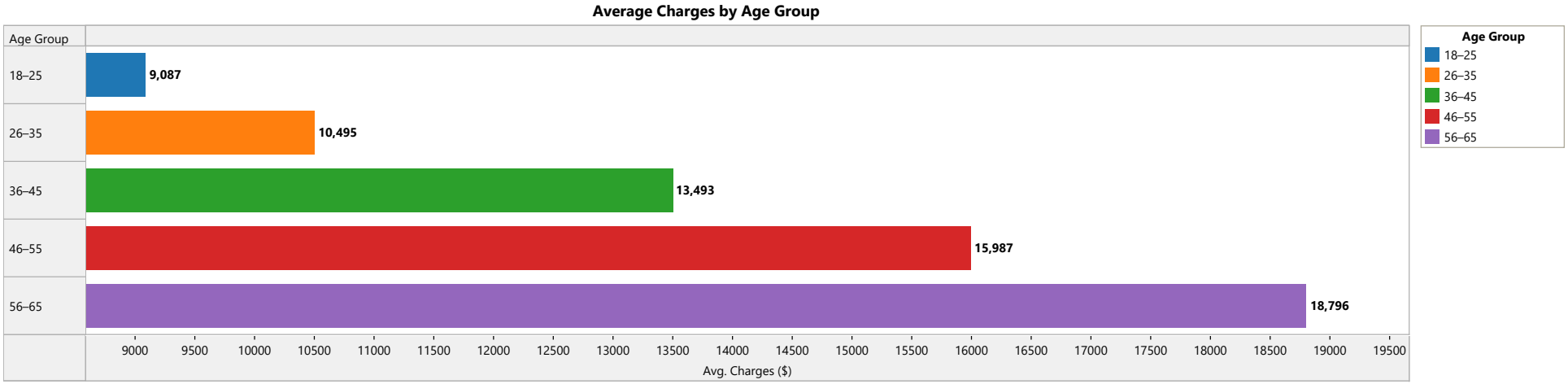
Average Charges by Gender × BMI Category

Sex	BMI Category			
	Underweight	Normal	Overweight	Obese
male	5,612	9,868	11,382	16,610
female	11,013	10,909	10,617	14,371

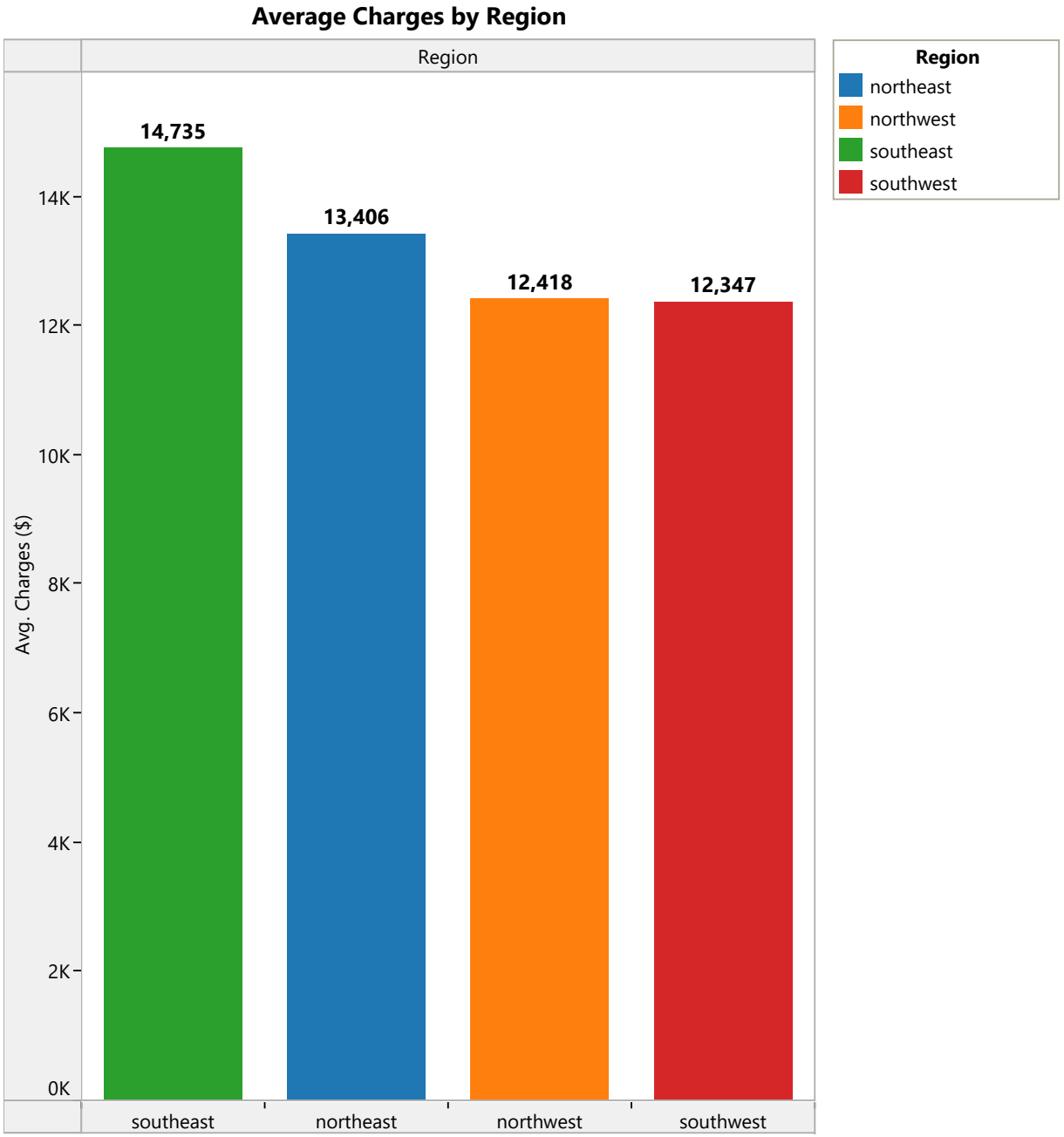
Avg. Charges

5,612 — 16,610

Underweight females show higher average charges than underweight males due to older age, higher smoking rates, and a $32k outlier claim. Obese males are the highest-cost demographic overall.



Average medical charges rise with age. Older age groups incur significantly higher costs, reflecting increased medical complexity and chronic conditions.



The Southeast region shows the highest average medical charges due to a higher concentration of smokers, obese individuals, and multiple high-cost claims. The Northwest and Southwest regions have lower averages because they contain fewer high-risk profiles and fewer extreme outlier charges.

Smoker × BMI Category Heatmap

BMI Category	Non-Smoker	Smoker
Underweight	5,533	18,810
Normal	7,686	19,942
Overweight	8,258	22,496
Obese	8,843	41,558

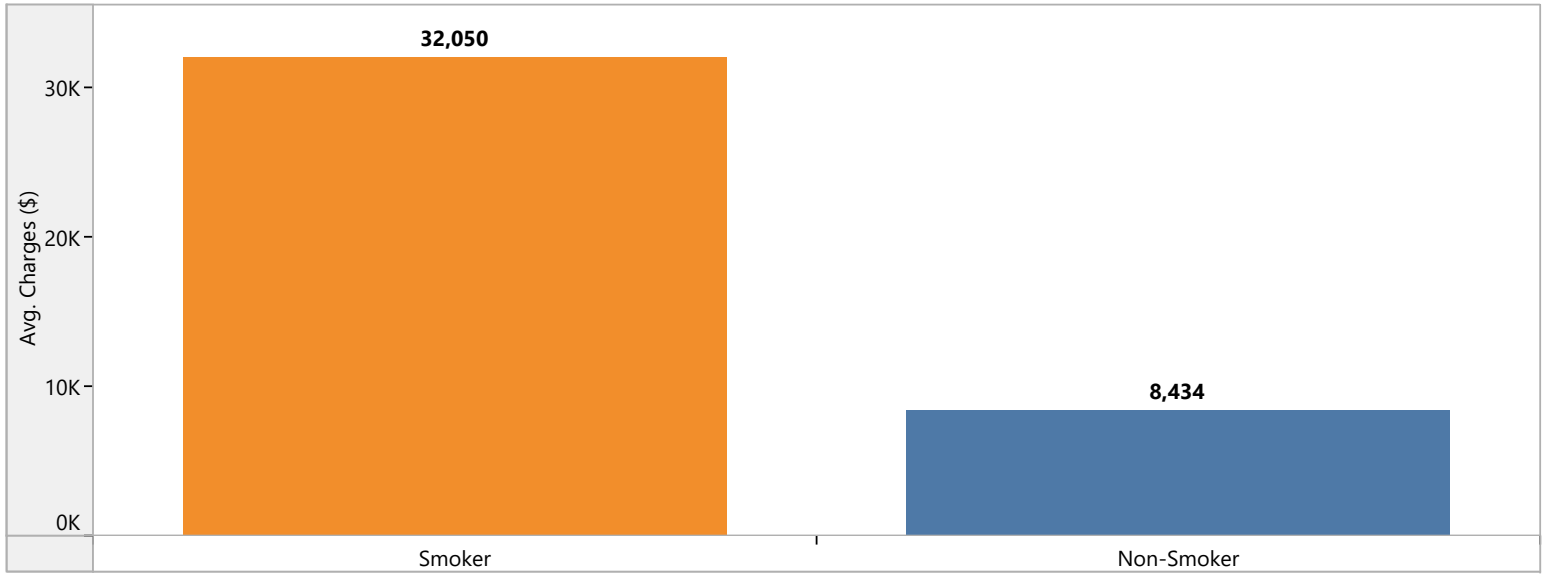
Avg. Charges

5,533 — 41,558

Smoker

- Non-Smoker
- Smoker

Average Charges by Smoker Status



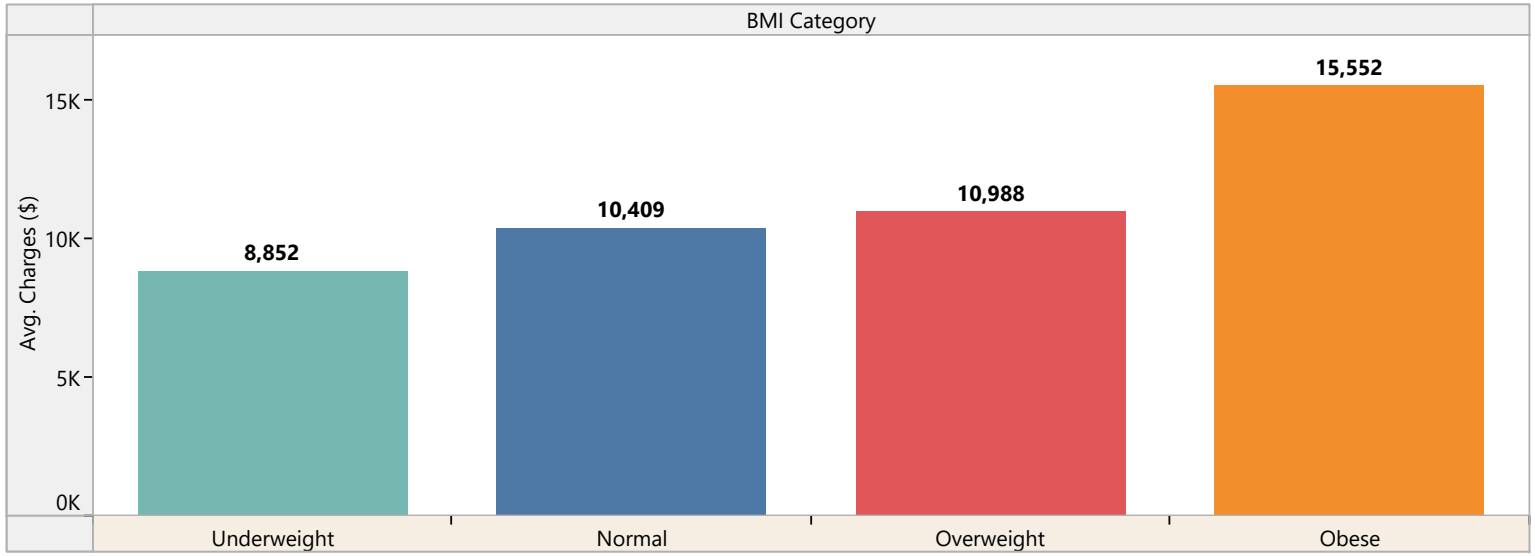
Average Charges by Gender × BMI Category

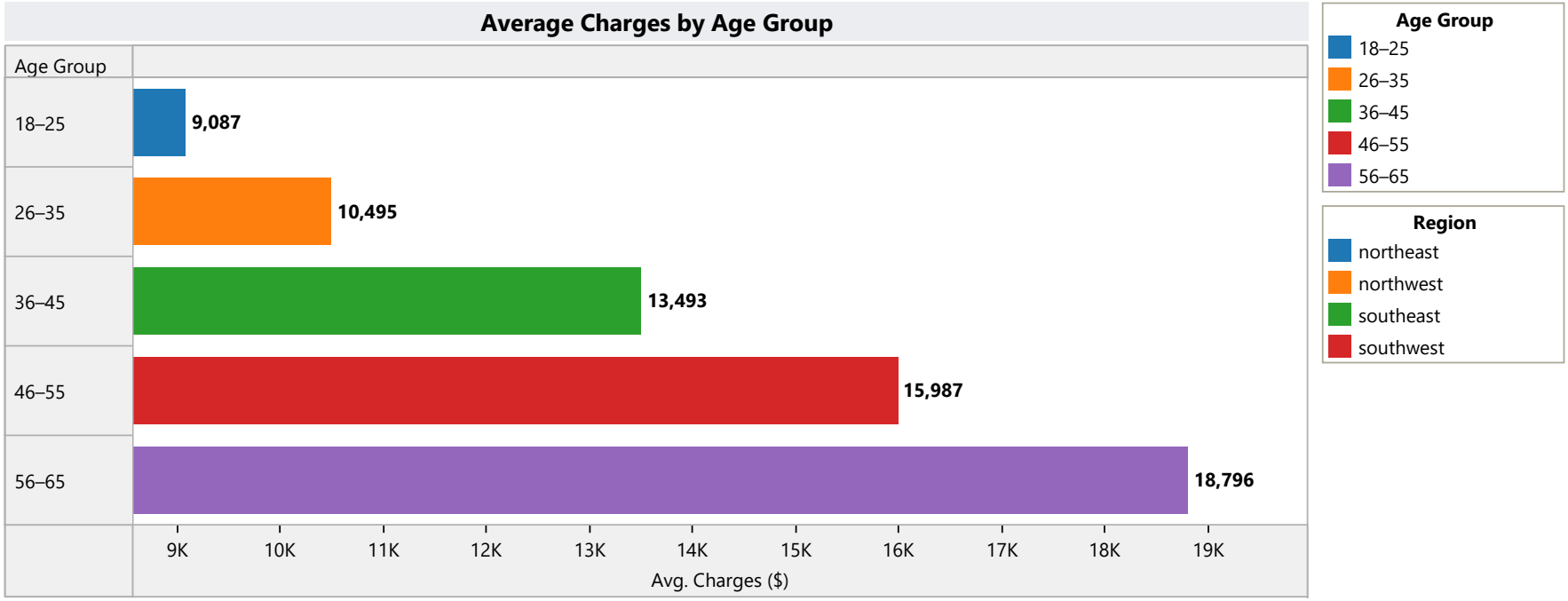
Sex	BMI Category: Underweight	Normal	Overweight	Obese
male	5,612	9,868	11,382	16,610
female	11,013	10,909	10,617	14,371

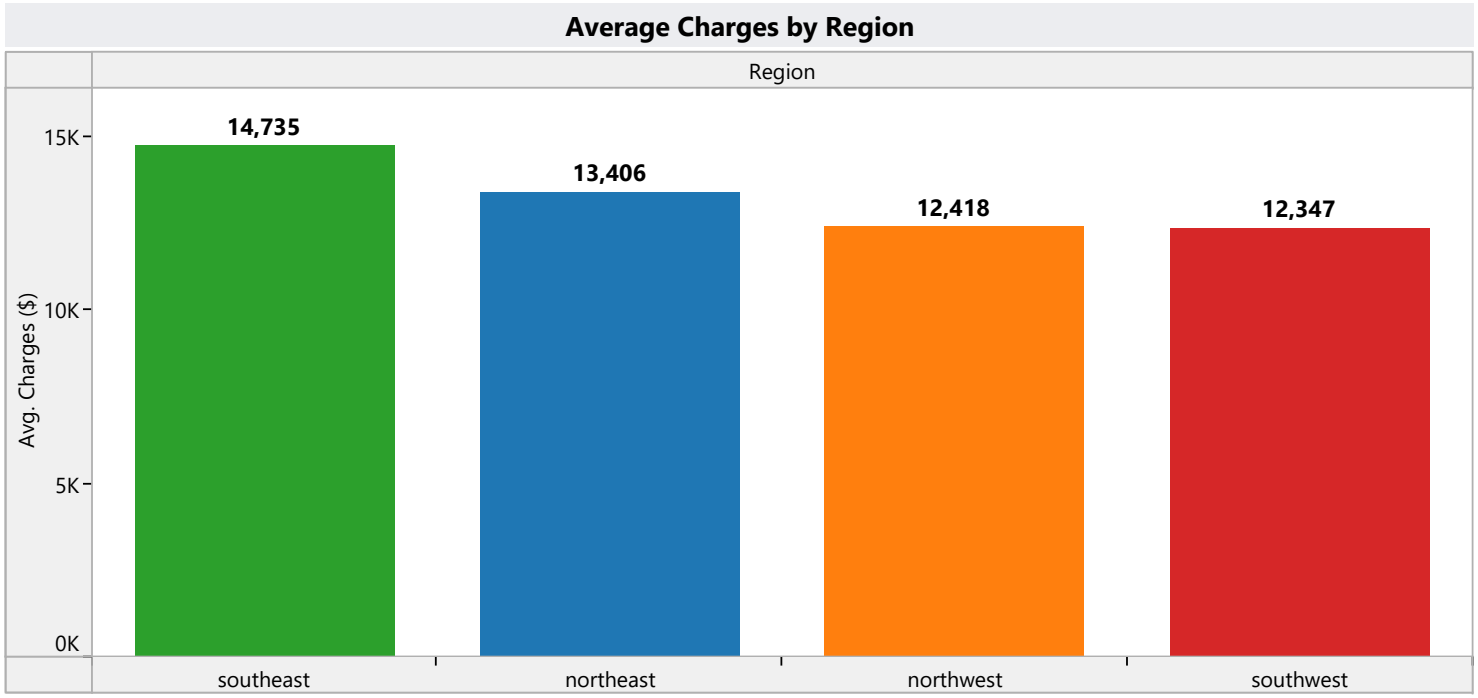
BMI Category
- Normal
- Obese
- Overweight
- Underweight

Avg. Charges
5,612 — 16,610

Average Charges by BMI Category




Average Charges by Age Group
Age Group
18–25
26–35
36–45
46–55
56–65
9,087
10,495
13,493
15,987
18,796
9K
10K
11K
12K
13K
14K
15K
16K
17K
18K
19K
Avg. Charges ($)
Age Group
18–25
26–35
36–45
46–55
56–65
Region
northeast
northwest
southeast
southwest


Average Charges by Region
Region
Avg. Charges ($)
15K
10K
5K
0K
14,735
13,406
12,418
12,347
southeast
northeast
northwest
southwest